tf.case Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/case

> *Create a case operation.*

Function Signatures

```
tf.case( pred_fn_pairs, default=None, exclusive=False,
strict=False, name='case' )
```

Code Examples

```
tf.case(
pred_fn_pairs,
default=None,
exclusive=False,
strict=False,
name='case'
)
```

```
tf.case(
pred_fn_pairs,
default=None,
exclusive=False,
strict=False,
name='case'
)
```

```
if (x < y) return 17;
else return 23;
```


```
if (x < y) return 17;
else return 23;
```


```
f1 = lambda: tf.constant(17)
f2 = lambda: tf.constant(23)
r = tf.case([(tf.less(x, y), f1)], default=f2)
```


```
f1 = lambda: tf.constant(17)
f2 = lambda: tf.constant(23)
r = tf.case([(tf.less(x, y), f1)], default=f2)
```


```
if (x < y && x > z) raise OpError("Only one predicate may evaluate
to True");
if (x < y) return 17;
else if (x > z) return 23;
else return -1;
```


```
if (x < y && x > z) raise OpError("Only one predicate may evaluate
to True");
if (x < y) return 17;
else if (x > z) return 23;
else return -1;
```

Documentation

Create a case operation.

Used in the notebooks

- Tutorial on Multi Armed Bandits in TF-Agents

See also tf.switch_case.

The pred_fn_pairs parameter is a list of pairs of size N.
Each pair contains a boolean scalar tensor and a python callable that creates the tensors to be returned if the boolean evaluates to True. default is a callable generating a list of tensors. All the callables in pred_fn_pairs as well as default (if provided) should return the same number and types of tensors.

If exclusive==True, all predicates are evaluated, and an exception is thrown if more than one of the predicates evaluates to True.
If exclusive==False, execution stops at the first predicate which evaluates to True, and the tensors generated by the corresponding function are returned immediately. If none of the predicates evaluate to True, this operation returns the tensors generated by default.

tf.case supports nested structures as implemented in tf.nest. All of the callables must return the same (possibly nested) value structure of lists, tuples, and/or named tuples. Singleton lists and tuples form the only exceptions to this: when returned by a callable, they are implicitly unpacked to single values. This behavior is disabled by passing strict=True.

Example 1:

Pseudocode:

Expressions:

Example 2:

Pseudocode:

Expressions:

Returns

Raises

v2 compatibility

pred_fn_pairs could be a dictionary in v1. However, tf.Tensor and tf.Variable are no longer hashable in v2, so cannot be used as a key for a dictionary. Please use a list or a tuple instead.